

COSMIC PLAYGROUND

A physics lab you can break. Guide for students and teachers.

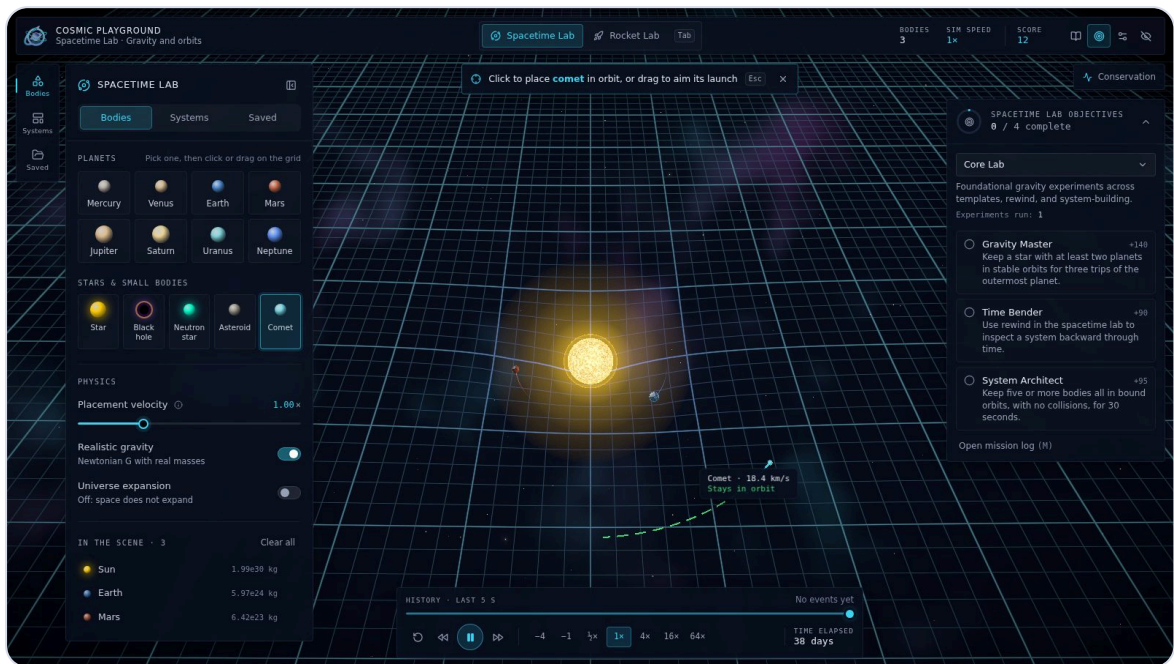


Textbooks show physics as finished answers. In Cosmic Playground you make a prediction, run the experiment, watch it succeed or fail, and work out why. Two labs: a **Spacetime Lab** for gravity and orbits, and a **Rocket Lab** for launching to orbit through real weather.

Runs in any modern browser. Nothing to install for students.

1. The Spacetime Lab

Every body pulls on every other body, just as in the real Solar System. Masses are real (the Sun is 1.989×10^{30} kg), one astronomical unit (AU) is 8 grid squares, and one year is the time an orbit at 1 AU takes. The grid bends under each mass so you can see the gravity well.



Placing a comet: before you click, the preview line shows where it will go and what will happen.

Placing bodies

1. Pick an object in the **Bodies** panel on the left: one of 8 real planets, a star, black hole, neutron star, asteroid or comet.
2. Move over the grid. The **preview line** shows the next 24 simulated seconds, worked out with the same physics as the simulation: **green** it stays in orbit, **amber** it escapes, **red** it hits something (and when).
3. **Click** to drop it into a circular orbit, or **press and drag** to throw it with your own speed and direction. **Esc** cancels.

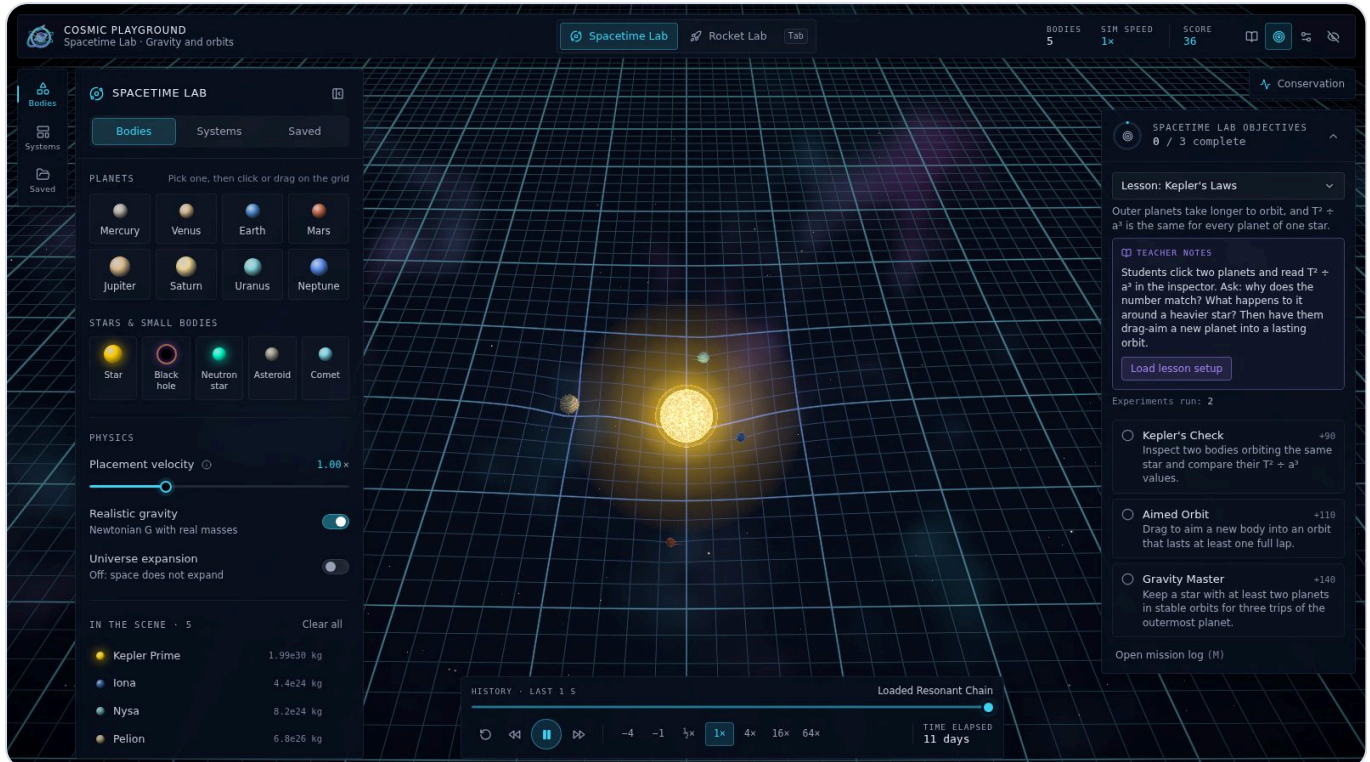
The **placement velocity** slider scales the orbital speed: below 1 the body falls inwards, above about 1.41 it escapes. Try it and check the preview.

What can happen

EVENT	WHEN
Merge	A slow collision: the bodies join, keeping their total momentum.
Bounce	Faster than the pair's escape speed: they rebound and lose energy as heat.
Shatter	Much faster: the smaller body breaks into fragments.
Swallowed	Anything crossing a black hole's event horizon.
Torn apart	A body that strays inside a black hole's Roche limit is stretched into a stream.

Each impact opens a short explanation with the momentum kept and the energy released, and leaves a marker on the history bar.

2. Measuring orbits



A teacher pack loaded: the Resonant Chain system with its lesson notes and objectives.

Body inspector

Click any body (in the scene or under *In the scene* on the left). The inspector shows what it orbits, its distance and speed, the size and shape of the orbit, how long one lap takes, its closest and farthest points, and $T^2 \div a^3$. For every planet of the same star that number comes out the same: Kepler's third law. A dashed line in the scene shows the orbit it is following.

Pin holds a body still while it keeps pulling on the others; **Follow** keeps the camera on it.

Time and conservation

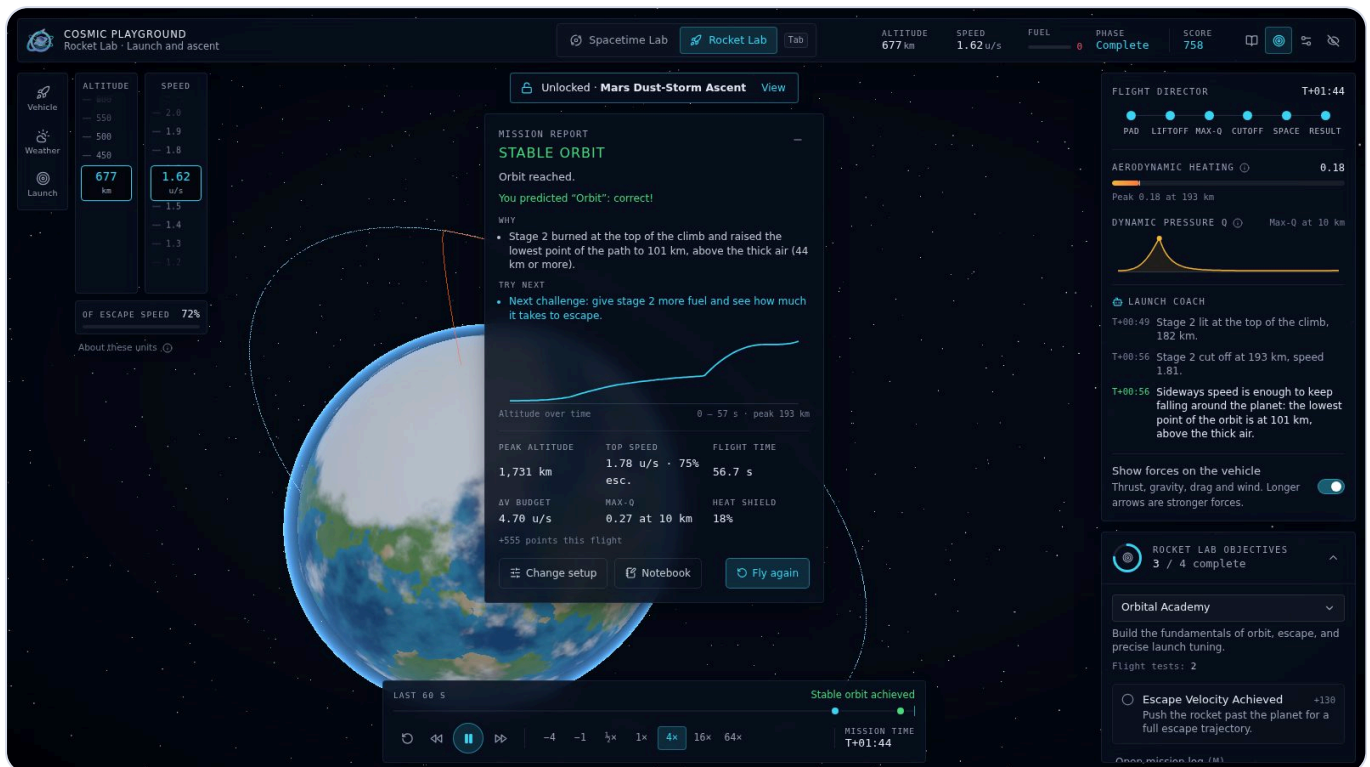
The time controls at the bottom pause and play (Space) and run from $-4\times$ (rewind) to $64\times$. At $64\times$ a year passes in under a second, with exactly the same result as at $1\times$. Drag the **history bar** to scrub back through the last 30 seconds; impact markers show when things happened, and absorbed bodies come back.

The **Conservation** button (top right) plots energy, momentum and angular momentum. They stay flat while only gravity acts, and jump at a collision, where energy turns into heat.

Arcade gravity (switch off *Realistic gravity*) is $4\times$ stronger with a speed cap: same shapes, faster play.

3. The Rocket Lab

Launch a rocket from a small planet and try to reach orbit. The flight follows real physics: gravity weakens with height, the air thins with altitude, drag depends on speed through the air (and the wind), and fuel burns away, making the rocket lighter. This is the rocket equation at work.



A two-stage launch in a stable orbit: the mission report, the flight director and the altitude and speed tapes.

Before launch

- **Launch setup** has three steps. **Vehicle:** thrust, burn time, fuel and dry mass, pitch-over angle and staging. **Weather:** air, wind, heating, temperature and pressure. **Launch:** scenario, gravity, planet size and pad tilt.
- **Weather:** 8 conditions. Each changes the values flown (shown next to the slider), and some bring hazards such as lightning or frozen seals.
- The **vehicle readout** shows the Δv budget (the total speed change the fuel can give), the thrust-to-weight ratio (it must be above 1 to lift off) and the speed an orbit needs.

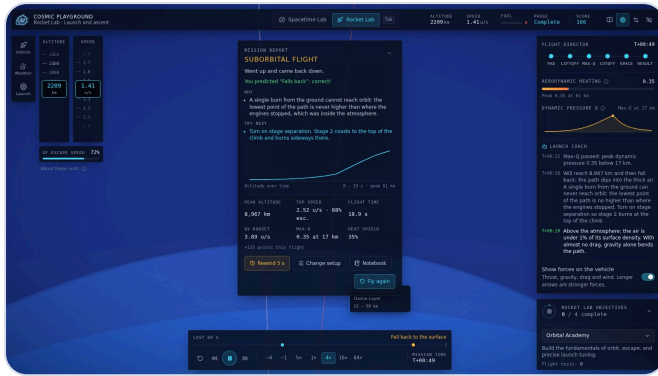
Why one stage is not enough

A rocket that burns all its fuel near the ground comes back down: the lowest point of its path can never be higher than where the engine stopped. Turn on **stage separation**. Stage 1 drops away when it is empty; stage 2 coasts to the top of the climb and burns sideways there, lifting the lowest point of the path above the air.

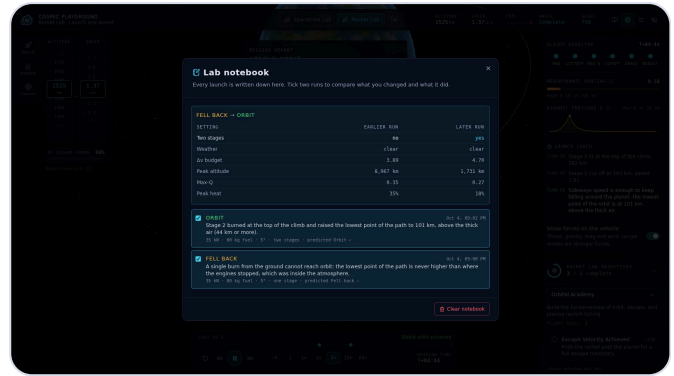
Outcomes

Orbit, escape, falls back, crash or burn-up, judged from the rocket's energy and the lowest point of its path. The verdict comes as soon as the outcome is certain, often before the top of the climb. Watch **Max-Q** (the hardest push of the air on the rocket) and the **heat shield**. Turn on *Show forces on the vehicle* to see thrust, gravity, drag and wind as arrows.

4. Predict, launch, explain



The debrief names the cause and one change to try.



The lab notebook compares two runs setting by setting.

1. **Predict.** Above the Ignite button, choose what you think will happen. You score points when you are right.
2. **Launch.** Ignite starts a three-second countdown (launches now). During the flight the **launch coach** reads the telemetry: Max-Q, heating, staging, and how much sideways speed an orbit still needs.
3. **Read the mission report.** What happened, why (with the numbers), and one specific change to try next. After a failure, **Rewind 5 s** plays the last moments back.
4. **Compare.** Every launch is written into the **Notebook** (in the mission report, or the notebook icon at the top of Launch setup). Tick two runs to see exactly which settings changed and what that did to the Δv , peak height, Max-Q and heating.

Change one thing at a time. If two settings change between runs, the notebook cannot tell you which one mattered.

Objectives

Objectives are checked against what really happens in the simulation, not just what you click. For example, *Gravity Master* needs planets that stay in orbit for three laps of the outermost one, and *Slingshot Expert* needs a flyby that really raises a comet's speed by 30%. There are 20 objectives in 11 courses. Pick a course in the Objectives panel on the right; the **Mission log** () lists them all. Points unlock a Gravity Slingshot system (500) and a Mars dust-storm ascent (750).

5. For teachers

TEACHER PACKS

Each teacher pack is one lesson: notes with questions to ask, three objectives, and a **Load lesson setup** button that puts the right system or rocket on every screen. Choose the pack in the course menu of the Objectives panel.

LESSON	LAB	BIG IDEA
Kepler's Laws	Spacetime	Farther planets are slower; $T^2 \div a^3$ is the same for every planet of one star.
Gravity Assists	Spacetime	A moving body can lend speed to a small one that swings past it.
Black Holes and Tides	Spacetime	Orbits around a black hole obey the same rules, until tides tear a body apart.
The Rocket Equation	Rocket	Fuel is heavy: doubling it does not double the speed; dropping empty tanks helps.
Weather and Launch Safety	Rocket	Wind, heat and thick air decide whether a launch is safe.

Sharing set-ups. Save a system (Spacetime Lab, *Saved* tab) or a rocket preset (Launch setup, *Saved setups*), then press **Export file** to download everything saved on that computer. Students press **Import file** to add it to theirs. Files are checked before anything is stored.

Settings and accessibility

Help and settings (top right, or ) holds **graphics quality** (use Low on older laptops and Chromebooks), **motion** (follows the device setting, or choose Reduced or Full) and **high contrast**.

KEY	ACTION
	Pause or play
 	Slower or faster; left of 0 rewinds
	Reset the current lab
	Stop placing a body, close the inspector or a dialog
  / 	Spacetime Lab / Rocket Lab. Tab switches labs only when no control is focused, so it still moves between controls.
  	Objectives, mission log, hide the interface
Drag / scroll	Turn the camera / zoom

Good to know

- Saved systems, presets, the notebook and settings are kept in the browser on that computer. Export a file to move them.
- The rocket's planet is small (radius around 50 km) so that orbits happen within a lesson; the physics is the same as for Earth.

- Very crowded systems cannot always run at 64x; the time controls then show the speed actually reached, rather than taking less accurate steps. Rewind covers the last 30 simulated seconds.